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MFCC-CNN ACOUSTIC SENSING ON ESP32 FOR REAL-TIME WATER LEAK DETECTION AND CLOUD NOTIFICATION

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INTRODUCTION

Water wastage caused by leaking taps is a common problem in homes, offices, schools, hotels, and public restrooms. Even a small drip from a faucet can waste thousands of liters of water annually. Most existing water leak detection systems rely on water flow sensors or moisture sensors, which may not effectively detect minor leaks at an early stage.

This project proposes an AI-powered acoustic leak detection system that uses sound analysis to identify water dripping from taps. By combining Deep Learning, IoT, Cloud Computing, and Mobile Notifications, the system can automatically detect leaks, alert users, and help conserve water.

PROBLEM STATEMENT

Small water leaks from taps often go unnoticed for long periods, resulting in:

- Water wastage
- Increased utility bills
- Environmental impact
- Maintenance issues

Traditional sensor-based systems may fail to detect small dripping leaks effectively.

Therefore, there is a need for an intelligent, real-time system capable of identifying water leaks at an early stage using sound recognition technology.

Project Objective

PRIMARY OBJECTIVE:

To develop an AI-based system that detects water leakage from taps using acoustic signals and automatically sends alerts to users.

SECONDARY OBJECTIVES :

- Reduce water wastage
- Provide real-time leak detection
- Generate instant notifications
- Estimate water loss
- Maintain leak history and analytics
- Improve water conservation efforts

Literature Review

S.No	Paper Title	Methodology Used	Findings	Limitations
1	**AI-based Acoustic Leak Detection in Water Distribution Systems (2022)**	Acoustic sensors with AI-based sound classification	Successfully detected water leaks using acoustic signals and machine learning techniques.	Focuses on large water distribution pipelines and does not provide household-level monitoring.
2	**Interpretable Deep Learning for Acoustic Leak Detection in Water Distribution Systems (2025)**	CNN-based Deep Learning Model	Achieved high accuracy in classifying leak sounds from pipeline systems.	Requires specialized equipment and is designed for industrial environments.
3	**Acoustic-Based Leak Detection and Localization Using Deep Learning(2025)**	Acoustic sensing with Deep Neural Networks	Detects leaks and estimates leak locations using sound analysis.	High implementation cost and complex deployment.

4	**IoT-Based Smart Water Monitoring and Leak Detection System(2025)**	Flow sensors integrated with IoT devices	Enables real-time monitoring of water consumption and leak detection.	Unable to effectively identify minor dripping leaks.
5	**Smart Water Leakage Detection Using Wireless Sensor Networks(2017)**	Wireless sensor network and flow monitoring	Provides continuous monitoring of water systems and leak identification.	Detection depends on water flow measurements rather than acoustic analysis.

Research Gap Analysis

Existing Systems	Identified Gap
Mainly focused on underground pipelines and industrial water networks.	Lack of solutions for household and bathroom tap leakage detection.
Depend on flow sensors and moisture sensors.	Small dripping leaks are difficult to detect at an early stage.
Limited real-time user notification mechanisms.	Need for instant mobile alerts and cloud monitoring.
Expensive infrastructure and deployment costs.	Requirement for a low-cost IoT-based solution.
Focus on leak detection only.	Need for water conservation analytics and monitoring.

Proposed Methodology

STEP 1: AUDIO COLLECTION

A microphone sensor continuously monitors sounds near the tap.

STEP 2: AUDIO PREPROCESSING

Noise removal and feature extraction techniques are applied to improve audio quality.

STEP 3: FEATURE EXTRACTION

MFCC (Mel Frequency Cepstral Coefficients) features are extracted from the audio signal.

STEP 4: CNN CLASSIFICATION

The CNN model classifies audio into:

1. Water Dripping
2. Running Water
3. Human Speech
4. Fan Noise
5. Background Noise

Proposed Methodology

STEP 5: LEAK DETECTION

If the system detects a dripping sound continuously for a predefined duration, it identifies a leak.

STEP 6: ALERT GENERATION

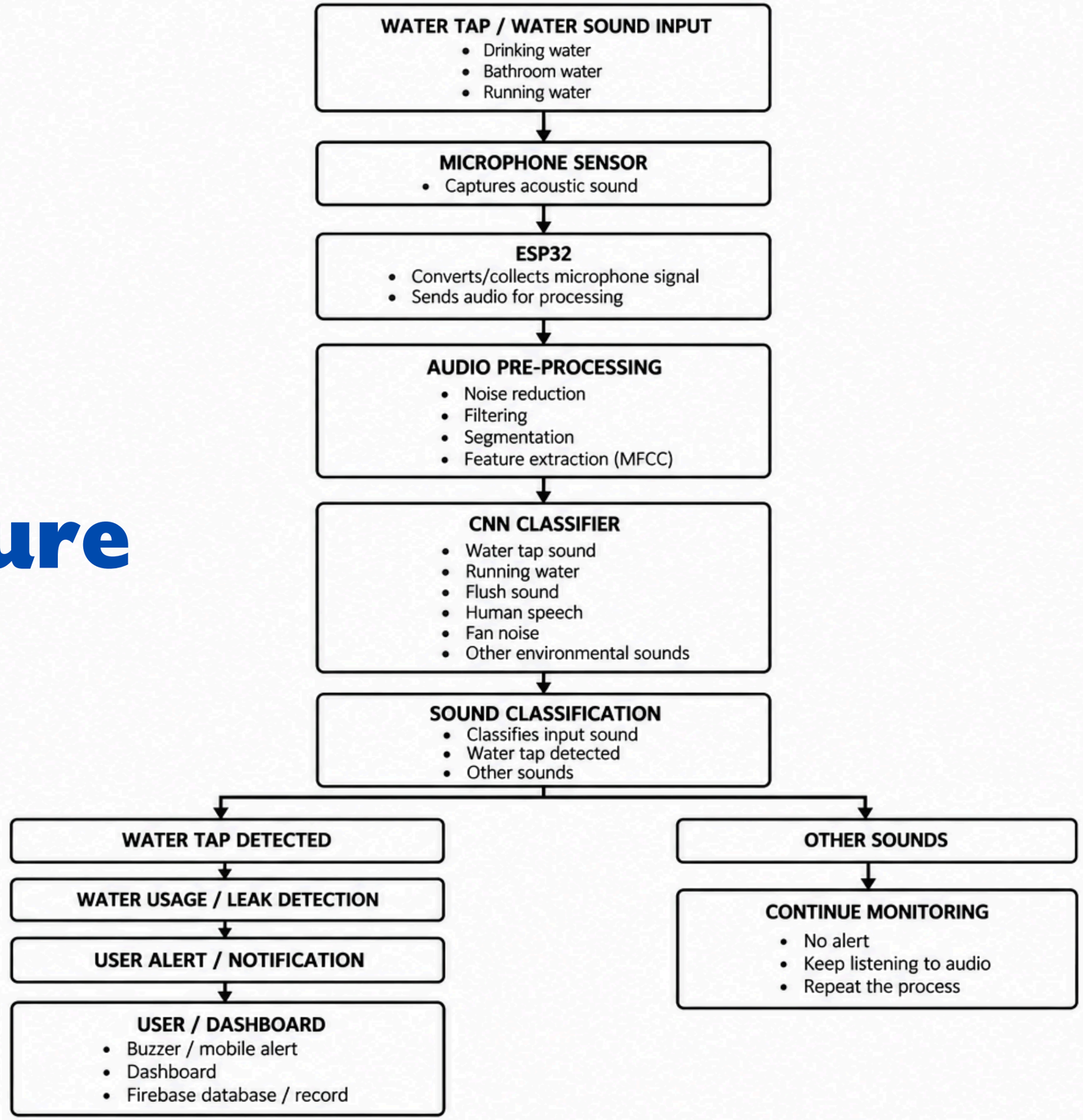
The system:

- Activates a buzzer
- Sends a mobile notification
- Updates cloud storage

STEP 7: DATA STORAGE

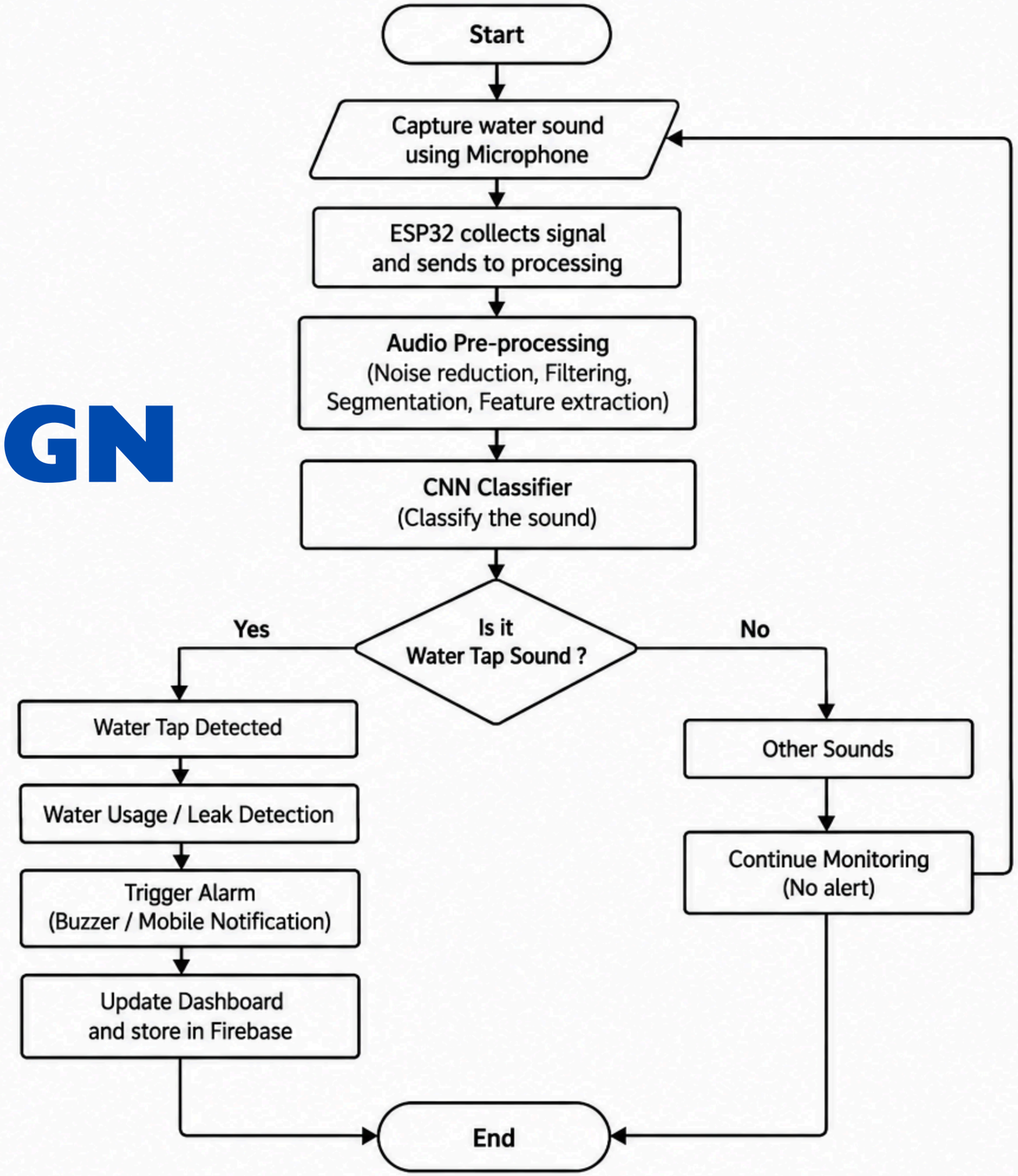
Leak information is stored in Firebase for future analysis.

System Architecture

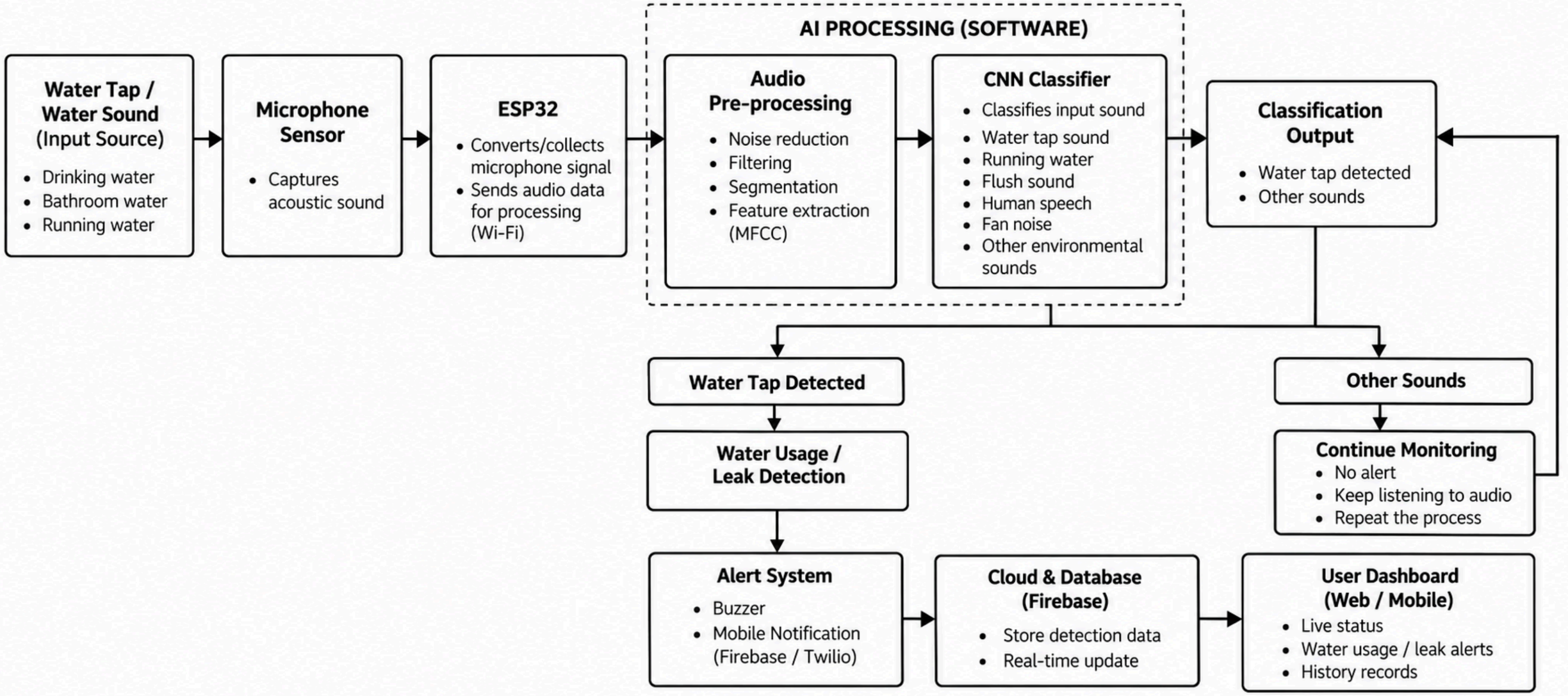


SYSTEM DESIGN

Flow Chart



BLOCK DIAGRAM



Leak Confirmation Algorithm

Step 1: Capture audio signal.

Step 2: Perform audio preprocessing.

Step 3: Extract MFCC features.

Step 4: Apply CNN classification.

Step 5: Check prediction confidence against threshold.

Step 6: Update leak confirmation counter.

Step 7: If leak persists for predefined duration, confirm leak.

Step 8: Trigger buzzer and notification.

Step 9: Store event in Firebase.

Step 10: Update dashboard and continue monitoring.

Module - Wise Implementation Progress

Module 1: Dataset Collection and Preparation

OBJECTIVE	To collect and organize acoustic data required for training the leak detection model.
IMPLEMENTATION	1) Collected audio samples from ESC-50 dataset. 2) Recorded custom sounds including water drip, fan noise, speech, and background noise. 3) Organized audio files into class-wise folders. 4) Converted and standardized audio files in WAV format.
PROGRESS	✅Dataset Collection Completed

Module - Wise Implementation Progress

Module 2: Audio Preprocessing

OBJECTIVE	To improve audio quality and prepare sound signals for feature extraction.
IMPLEMENTATION	1) Loaded audio using Librosa. 2) Removed unwanted noise. 3) Normalized audio amplitude. 4) Resampled audio signals to 22050 Hz. 5) Segmented long audio recordings.
PROGRESS	✅ Completed

Module - Wise Implementation Progress

Module 3: Feature Extraction

OBJECTIVE	To convert audio signals into numerical features suitable for AI model training.
IMPLEMENTATION	1) Applied MFCC feature extraction. 2) Extracted 40 MFCC coefficients from each audio sample. 3) Generated feature vectors for classification.
PROGRESS	✅ Completed

Module - Wise Implementation Progress

Module 4: CNN Model Development

OBJECTIVE	To classify water leakage sounds using a deep learning model.
IMPLEMENTATION	1) Designed CNN architecture. 2) Defined convolution and pooling layers. 3) Prepared training and testing datasets. 4) Configured model training parameters.
PROGRESS	✅ Completed

Module - Wise Implementation Progress

Module 5: Hardware Integration

OBJECTIVE	To acquire acoustic signals and generate real-time leak alerts.
IMPLEMENTATION	1) Selected ESP32 controller. 2) Integrated INMP441 microphone sensor. 3) Connected buzzer for local alerts. 4) Designed hardware communication architecture.
PROGRESS	🟡 In Progress

Module - Wise Implementation Progress

Module 6: Firebase Integration

OBJECTIVE To store leak events and support remote monitoring.

IMPLEMENTATION

1) Created Firebase project.
2) Designed cloud database structure.
3) Configured real-time data synchronization.
4) Prepared leak event storage schema.

PROGRESS 🟡 In Progress

Module - Wise Implementation Progress

Module 7: Dashboard Development

OBJECTIVE	To provide real-time visualization and monitoring of leak events.
IMPLEMENTATION	1) Designed Login and Registration pages. 2) Developed Dashboard interface. 3) Created Leak History and Analytics modules. 4) Integrated monitoring features.
PROGRESS	✅ Completed

Module - Wise Implementation Progress

Module 8: Notification System

OBJECTIVE To provide immediate alerts when leakage is detected.

IMPLEMENTATION

1) Defined leak confirmation logic.
2) Configured buzzer alarm mechanism.
3) Designed notification workflow.
4) Planned dashboard alert integration.

PROGRESS 🟡 In Progress

Prototype / Screenshots / Working Demonstration

Prototype Developed :

The prototype of the proposed system consists of both hardware and software components.

Hardware Prototype :

1. ESP32 Development Board
2. INMP441 Microphone Sensor
3. Buzzer Alert System
4. Power Supply Unit

Software Prototype :

1. Audio Dataset Collection
2. Audio Preprocessing Module
3. MFCC Feature Extraction Module
4. CNN Model Development
5. Firebase Database Setup
6. Dashboard Interface Design

Working Demonstration :

1. Microphone captures sound signals.
2. Audio is processed and converted into MFCC features.
3. CNN model analyzes the sound pattern.
4. Leak sound is identified.
5. Buzzer alert is activated.
6. Leak information is sent to Firebase.
7. Dashboard displays leak status and notifications.

IMPLEMENTATION

```
_rate: 2.5000e-04
2026-09-27 14:41:11,670 INFO src.training.trainer: CNN model report:
                precision    recall  f1-score   support

Background_Noise       1.00      1.00      1.00        10
       Fan_Noise       0.82      1.00      0.90         9
     Flush_Sound       1.00      1.00      1.00        10
    Human_Speech       1.00      1.00      1.00        10
   Running_Water       1.00      1.00      1.00        10
      Water_Drip       1.00      0.88      0.94        17

        accuracy                           0.97        66
       macro avg       0.97      0.98      0.97        66
    weighted avg       0.98      0.97      0.97        66

2026-09-27 14:41:11,681 INFO src.training.trainer: Evaluation complete. MFCC accuracy=0.9242, CNN accuracy=0.9697
{
  "mfcc": {
    "accuracy": 0.9242424242424242,
    "precision_macro": 0.9458333333333333,
    "recall_macro": 0.9198257080610022,
    "f1_macro": 0.9295471942530766
  },
  "cnn": {
    "accuracy": 0.9696969696969697,
    "precision_macro": 0.9696969696969697,
    "recall_macro": 0.9803921568627452,
    "f1_macro": 0.9729166666666668
  }
}
```

CNN MODEL REPORT

Class	Precision	Recall	F1
Background Noise	1.00	1.00	1.00
Fan Noise	0.82	1.00	0.90
Flush Sound	1.00	1.00	1.00
Human Speech	1.00	1.00	1.00
Running Water	1.00	1.00	1.00
Water Drip	1.00	0.88	0.94

Interpretation

- Water Drip Precision = **1.00**
  - Every leak prediction was correct.
  - No false leak alarms.
- Water Drip Recall = **0.88**
  - 88% of actual leak sounds detected.
- F1 Score = **0.94**
  - Excellent balance.

Overall Accuracy : 96.97

MFCC MODEL REPORT

Class	Precision	Recall	F1
Background Noise	1.00	0.80	0.89
Fan Noise	0.88	0.78	0.82
Flush Sound	1.00	1.00	1.00
Human Speech	1.00	1.00	1.00
Running Water	1.00	1.00	1.00
Water Drip	0.80	0.94	0.86

Interpretation

- Water Drip Recall = **0.94**
  - 94% of actual leak sounds were detected.
- Water Drip Precision = **0.80**
  - Some non-leak sounds were wrongly predicted as leaks.
- Overall Accuracy = **92.42%**

MFCC model performs well but has a few false detections.

MFCC VS CNN MODEL

Why Train Both?

In machine learning projects, we train multiple models and compare them.

Metric	MFCC Model	CNN Model
Accuracy	92.42%	96.97%
Precision	94.58%	96.97%
Recall	91.98%	98.04%
F1 Score	92.95%	97.29%

Dataset collection

Name	Date modified	Type	Size
Background_Noise	27-09-2026 12:53	File folder	
Fan_Noise	27-09-2026 12:52	File folder	
Flush_Sound	27-09-2026 12:51	File folder	
Human_Speech	27-09-2026 12:50	File folder	
Running_Water	27-09-2026 12:50	File folder	
Water_Drip	27-09-2026 12:49	File folder	
.gitkeep	25-09-2026 19:55	GITKEEP File	0 KB

Dashboard

AquaSense

Sign in | Open dashboard

Live acoustic monitoring

Smart Water Leak Monitoring System

An AI-based acoustic water leakage detection platform that listens to municipal pipelines, classifies leak signatures in real time and dispatches severity-graded alerts before water loss escalates.

Launch live demo | Create account

- Acoustic sensor fleet management
- Severity-graded leak alerts
- Zone-wise water loss analytics
- Board-ready exportable reports

Acoustic waveform view



Sign in to stream live decibel and frequency readings from your own sensor nodes.

Platform capabilities

Everything required for an end-to-end leak detection deployment — from edge acoustic sensing to board-ready reporting.

AI acoustic classification

Deep-learning model separates leak signatures from ambient pipeline noise.

Real-time monitoring

Live dB and frequency streams from every acoustic node with threshold breach alerts.

Instant notifications

Severity-graded alerts routed to the right field engineer within seconds.

Analytics & reports

Zone-wise trends, water saved and exportable PDF/Excel reports for review boards.

Device health

Battery, signal and firmware telemetry for the entire sensor fleet.

Role-based access

Admin, engineer and viewer roles backed by secure cloud authentication.

Leak Detection

Search devices, zones, leak IDs...

Live stream

BP BHUVANENDIRAN P

Audio Leak Detection

Upload a pipe or sensor recording — AI will tell you whether it sounds like a leak.

Upload recording

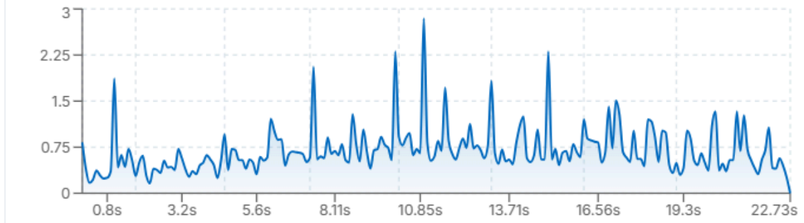
WAV, MP3, M4A, OGG, AAC, FLAC or WEBM · up to 10 MB

Click or drop an audio file here

freesound_community-039645_water-drops-67983.wav 2.09 MB

0:22 / 0:22

Sound level over time



Detect leak

Result

AI acoustic classification

Leak detected

Medium Severity

Confidence 92%

Distinct, rapid dripping and splashing sounds indicating an active liquid leak, likely from a pipe or faucet.

No Leak Detection

Audio Leak Detection

Upload a pipe or sensor recording — AI will tell you whether it sounds like a leak.

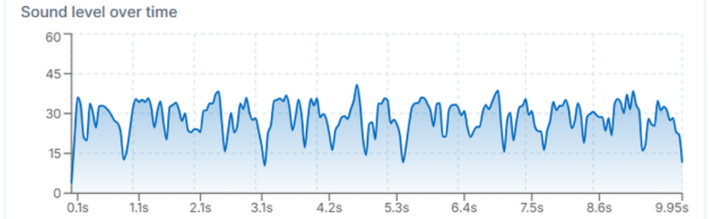
Upload recording

WAV, MP3, M4A, OGG, AAC, FLAC or WEBM · up to 10 MB

Click or drop an audio file here

abirkhan006-10-seconds-of-savage-vibe-591825.mp3 0.31 MB

0:00 / 0:10



Detect leak

Result

AI acoustic classification

No leak detected

Confidence 98%

The recording contains rhythmic electronic music with synthesized beats rather than any continuous hissing, rushing, or dripping sounds characteristic of a water pipe leak.

Software Requirements

Software	Purpose
Python	AI Model Development
TensorFlow	Deep Learning
Librosa	Audio Processing
Arduino IDE	ESP32 Programming
Firebase	Cloud Database
React JS	Dashboard Development
Visual Studio Code	Development Environment

Hardware Requirements

Component	Quantity
ESP32 Development Board	1
INMP441 MEMS Microphone	1
Buzzer	1
LEDs	2
Breadboard	1
Jumper Wires	As Required
Power Supply	1

Final Results & Performance Analysis

1. The proposed AI-Based Acoustic Water Leak Detection System was successfully developed using MFCC feature extraction and a CNN model for acoustic sound classification.
2. The model was trained and tested using multiple sound categories including Water Drip, Running Water, Human Speech, Fan Noise, Flush Sound, and Background Noise.
3. The system achieved an Overall Accuracy of 96.97%, demonstrating excellent classification performance.
4. The Water Drip class achieved a Precision of 1.00, indicating that all predicted leak events were correctly identified without false leak alarms.
5. The model achieved a Recall of 0.88 and an F1-Score of 0.94 for leak detection, showing reliable identification of actual leakage sounds.
6. The system successfully differentiated leak sounds from environmental noises and provided accurate detection results.
7. Experimental results confirm that the proposed solution is effective, reliable, and suitable for real-time water leak detection and water conservation applications.

Key Performance Metrics

Metric	Value
Overall Accuracy	96.97%
Water Drip Precision	1.00
Water Drip Recall	0.88
Water Drip F1-Score	0.94

Challenges Faced

CHALLENGE 1: LIMITED WATER LEAK DATASET

Issue: Publicly available water drip datasets are limited.
Solution: Combined ESC-50 dataset with custom audio recordings.

Challenge 2: Similarity Between Leak and Water Flow Sounds

Issue:
Leak sounds and normal water flow sounds have similar acoustic characteristics.
Solution:
Used MFCC feature extraction and CNN-based classification to identify unique sound patterns.

Challenge 3 : Hardware-Software Integration

Issue: Integration between ESP32, AI model, Firebase, and dashboard is still under development.
Solution: Adopted a modular architecture for easier implementation and testing.

References

1. Nam, Y. W., Arai, Y., Kunizane, T., & Koizumi, A. (2021). Water Leak Detection Based on Convolutional Neural Network Using Actual Leak Sounds and the Hold-Out Method. Water Supply, 21(7), 3477–3485. IWA Publishing
2. AI-based Acoustic Leak Detection in Water Distribution Systems (2022)
3. Interpretable Deep Learning for Acoustic Leak Detection in Water Distribution Systems (2025)
4. Rahman, M. H., Ahmed, S., & Islam, T. (2023). An Intelligent IoT and ML-Based Water Leakage Detection System. IEEE Access. The study combined acoustic sensing, machine learning, and IoT-based alert mechanisms for leak detection. IEEE Xplore
5. Zhang, Y. et al. (2025). Leak Detection of Underground Water Pipelines Using Acoustic Feature Extraction. IEEE Internet of Things Journal. The work used MFCC features and deep learning for accurate acoustic leak detection. IEEE Xplore
6. A. Kumar and P. Sharma, "Acoustic Signal Processing Techniques for Water Leak Detection in Smart Buildings," International Journal of Smart Systems, vol. 18, no. 4, pp. 112–120, 2024.
7. ESC-50 Dataset – Environmental Sound Classification Dataset. Used for collecting and classifying environmental audio samples such as speech, fan noise, and background sounds.

Gantt Chart

Task	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8
Problem Identification & Planning	■■■							
Literature Review & Research Gap Analysis	■■■	■■■						
Requirement Analysis & System Design		■■■	■■■					
Dataset Collection & Preparation			■■■	■■■				
Audio Preprocessing & MFCC Feature Extraction				■■■	■■■			
CNN Model Development & Training				■■■	■■■	■■■		
Dashboard Development (Web Application)					■■■	■■■	■■■	
Testing, Performance Analysis & Documentation						■■■	■■■	■■■

THANK YOU